

Leon Chlon

leochlon@gmail.com | linkedin.com/in/leochlon | github.com/leochlon | scholar.google.com/citations?user=-gWbKU0AAAAJ

PROFILE

AI Research Lead at PhysicsX, building in-context learning models for physics; recently Visiting Fellow at the University of Oxford (previously PhD, Cambridge University; Postdoc, Harvard/MIT), research into production systems serving 100M+ users; first-author ICML 2026 publication on LLM hallucination and creator of open-source AI-assurance tooling with industry integrations. Experienced in leading multidisciplinary teams & cross-functional workflows.

EDUCATION

Harvard Medical School / MIT - Postdoctoral Fellow (Machine Learning)	2018
University of Cambridge - Ph.D. (Machine Learning)	2017
University of Cambridge - M.Phil. (Theoretical Physics)	2014

EXPERIENCE

PhysicsX **September 2026 - Present**
AI Research Lead, Fundamentals

- Build in-context learning models that predict surface fields such as pressure from a handful of example simulations, without retraining, so they learn from their examples rather than memorise geometry.
- Developed the Gaussian SDPA now used across the team, which biases attention towards physically nearby surface points; added information-preserving normalisation, adapters and training-time augmentations.
- Pretrain geometry and field encoders with masked-reconstruction and distance-field objectives; build evaluations that separate real in-context learning from shortcuts (invariance audits, counterfactual probes, DrivAerML).

University of Oxford **March 2025 - September 2026**
AI Research Lead, Visiting Fellow

- Led AI research within Oxford's Torr Vision Group, producing an ICML 2026 paper; secured \$300K in backing from Microsoft, Google, HP and NVIDIA to fund the programme.
- Built partnerships across MIT, Cambridge and Oxford, co-supervising joint PhD and laboratory projects.
- Co-designed INT4-compression CUDA kernels (Integrations with Unsloth), cutting quantization penalty by ~9%; extended Triton kernels to Blackwell GB10 GPUs for 6.1× acceleration vs PyTorch.
- Built and released Mezzanine, a model-distillation toolkit adopted by teams at DeepMind, CERN, DAMTP and CRUKCI.

Selected Industry Engagements — World Bank Group, Apple, TikTok **August 2023 - March 2025**

Applied ML across ranking, forecasting and large-scale inference. Led AI reliability discussions with PwC, McKinsey, Novartis and government policymakers including SDAIA (Saudi Arabia).

- World Bank Group: fine-tuned LLaMA-2 70B (LoRA) across 300k socioeconomic indicators, cutting forecast error (sMAPE) by 35%; coordinated adoption with DECDG and DECPG team leads.
- Apple: led a cross-functional upgrade of a production recommender system; directed data engineering on monitored DAG pipelines (drift detection, latency alarms) and delivered a state-of-the-art MMoE ranker.
- TikTok: Bayesian-optimised retraining of a 500M+ DAU content-ranking system (Ax, BoTorch, MLflow, Ray), yielding double-digit gains in cross-platform sharing.
- Held AI reliability consultations with senior stakeholders at PwC, NVIDIA, Novartis and government policymakers including SDAIA (Saudi Arabia).

Tailor Bio (University of Cambridge spinout) **December 2022 - August 2023**
Founding AI Engineer

Leon Chlon

leochlon@gmail.com | linkedin.com/in/leochlon | github.com/leochlon

- Designed and deployed a drug-discovery AI stack on AWS (PyTorch, RDKit, Neo4j), training GNN foundation models with sparse matrix operators and gradient checkpointing.
- Led the founding MLE team and coordinated validation with wet-lab partners; presented three AI-discovered compounds to the board and investors, supporting a successful Series B raise.

Uber Technologies (Careem)

January 2022 – August 2022

Lead ML Engineer – Dynamic Pricing

- Redesigned marketplace pricing for 100M+ users using algorithmic game theory, achieving a 2.7× efficiency improvement over the legacy system and \$2M+ revenue impact.
- Co-designed feature-engineering DAGs with Data Platform, Product and Risk verticals; deployed with graceful degradation, automatic deprecation, canary rollouts and shadow testing (Kubernetes/Istio, MLflow).

Meta

October 2020 – January 2022

Senior Research Scientist – AI Safety

- Introduced Bayesian reinforcement-learning tools for crash detection across Instagram, WhatsApp and Facebook (PyMC3), achieving a 10-15% improvement in cross-platform detection.
- Improved hate-speech detection recall by 15% at billion-post scale (FAISS/HNSW) while maintaining precision.
- Built PySpark pipelines joining Hadoop/Hive data with real-time analytics; developed Prophet-based health-degradation warnings.

McKinsey & Company

November 2018 – June 2020

Senior Data Scientist

- Designed gradient-boosted credit-risk models for Tier 1 banks covering \$100B+ exposures, delivering a >40% improvement in early-warning detection for an Asia-based client.
- Presented Basel III/IFRS 9-compliant strategies to C-suite executives at three major financial institutions.

SELECT PUBLICATIONS & OPEN SOURCE

- [Predictable Compression Failures: Order Sensitivity and Information Budgeting for Evidence-Grounded Binary Adjudication](#) ([arXiv:2509.11208](#)) — First Author, ICML 2026.
- [Exact Finite Attention Responses From RoPE Derivatives](#) ([arXiv:2609.14127](#)) — PI and senior author, with student co-authors and Gregory Gutin (Royal Holloway); under review at ICLR 2027.
- [Robot World Models Are Not Invariant to How the Actions Are Written](#) ([arXiv:2609.23252](#)) — PI and senior author; under review at ICLR 2027.
- [LLMs are Bayesian, in Expectation, not in Realization](#) ([arXiv:2507.11768](#)) — arXiv preprint, 2025.
- [HallBayes](#), a model-agnostic post-training harness for AI Alignment and LLM hallucinations; **1.7k GitHub stars**, 10,000+ industry integrations including DeepSet and PwC. Featured at **NVIDIA GTC 2026**.
- [NTKMirror](#), a frozen-model activation-control framework for lightweight in-context learning and finetuning; 338 stars, 48 forks.
- Core contributions to DeepMind Optax and HuggingFace, including a 4x speedup in SAM image processing.

TEACHING & COMMUNITY

- Teaching Assistant, MIT 9.073/HST.460 Statistics for Neuroscience Research (Prof. Emery Brown); guest teaching, ES 201/APMTH 231 Decision Theory, Harvard SEAS (Prof. Demba Ba).
- Research mentorship of PhD and Masters students (MIT, Harvard), leading to co-authored publications. As PI, supervised three student researchers: two are now PhD students at UCL and the University of Barcelona, and the third is finalising PhD applications. EWOR Fellowship (0.1% acceptance rate).
- Invited talks on AI safety and production ML (Imperial College London, AUB, Sharjah); teach ML to a 180k+ audience on LinkedIn, Instagram and TikTok.